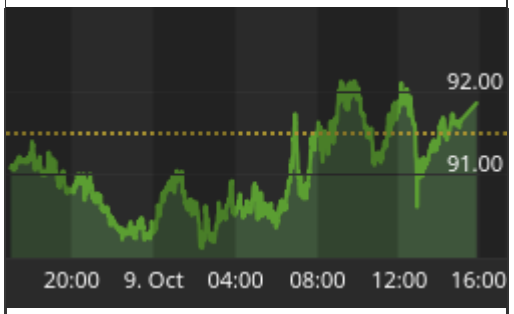
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Iran War Energy Shock Puts Hydrogen Back on the Table

By Haley Zaremba - Oct 08, 2026, 4:00 PM CDT

- Oregon State researchers built a sulfur-based photocatalyst that splits water into hydrogen using sunlight, with no expensive metal catalyst required.
- A process that swaps water oxidation for waste sugars has pushed green hydrogen to $1.54 per kilogram, roughly level with hydrog made from natural gas.

- Natural “white” hydrogen could skip production altogether, but nobody knows yet how much of it is recoverable or what it will cost to pull out of the ground.



This year’s global energy crisis has revived international interest in hydrogen’s potential to replace fossil fuels in a wide range of industrial applications, from steelmaking to shipping. Green hydrogen is still massively expensive to produce, reducing its potential for commercial scaling, while gray and blue hydrogen relies on the same fossil fuels that many global markets are trying to move away from. But scientific breakthroughs are piling up to expand the hydrogen rainbow in ways that could sidestep these pitfalls and potentially decarbonize and disrupt some of the world’s hardest-to-abate sectors.

Hydrogen holds massive potential for replacing fossil fuels in manufacturing and transportation because the element can be combusted at high temperatures like thermal coal or heavy fuel oil. But unlike those particularly high-emitting fossil fuels, hydrogen leaves behind

nothing but water vapor when it burns. As a result, green hydrogen – or that which is made using renewable energies alone – has long been touted as an invaluable tool for the global clean energy transition and decarbonization pathways.

However, green hydrogen ambitions have remained elusive. The process of splitting hydrogen from water is costly and energy-intensive, and in many cases producing green hydrogen is an inefficient use of clean energy that could be more effectively consumed directly. Indeed, a 2022 report by the International Renewable Energy Agency (IRENA) warned against the "indiscriminate use of hydrogen" and suggested that policymakers should instead weigh competing priorities carefully and consider that extensive use of green hydrogen "may not be in line with the requirements of a decarbonised world."

But a groundswell of funding into hydrogen research on the heels of the United States' and Israel's war in Iran has arrived at the right time for a number of scientific projects seeking to find alternative ways of sourcing hydrogen while maintaining a low carbon footprint.

Just this week, a team of scientists at Oregon State University published a paper elucidating a new method for deriving green hydrogen from water using sunlight alone, through the application of a sulfur-based chemistry. Compared to standard methods of splitting hydrogen from water using electricity, "Photocatalysis is a much more direct approach, as it uses a material in its high-energy state activated by sunlight to bring about the same reaction and at a faster rate," according to a summary from Interesting Engineering.

"Our work provides a blueprint for designing better materials that can bring down the cost of green hydrogen," said Oregon State's Kyriakos Stylianou in a press release. "By changing the metal while keeping the rest of the material essentially the same, we discovered why some versions of the MOF work much better than others. These findings provide new design rules for creating more effective materials for solar fuel production."

This discovery comes on the heels of a separate breakthrough from MIT, which involved the development of an electrochemical process that can pull high-purity hydrogen from ammonia at a fraction of the energy cost of traditional "cracking," which requires temperatures above 500 degrees Celsius.

And a January study out of China found yet another way to derive green hydrogen more cheaply and efficiently by using sugars from agricultural waste in place of oxygen in the hydrogen splitting process, lowering the cost of green hydrogen to $1.54 per kilogram, a price which would finally make the resource cost-competitive with natural gas.

But perhaps the biggest and buzziest development in the hydrogen sector this year is the promise of "white hydrogen," also known as geologic hydrogen. This approach does not seek to create hydrogen at all, but rather to harvest it from where it naturally occurs within the

Earth’ s crust. The technology to unlock this hydrogen from the rocks where it has been stored for millions of years is still under development, but the potential is massive and has already received attention from the likes of the United States Geological Survey, which estimates that the ground beneath our feet contains energy equivalent to roughly 170,000 years of present-day global oil consumption. The question is how much of that is recoverable, and at what price point.

What is clear is that, after years of plateau, clean hydrogen hype is back, and with good reason. Scientific evidence is piling up to suggest that a clean hydrogen takeover could take a number of different forms, but that one of them is eventually likely to succeed, changing the entire global energy landscape in the process.

By Haley Zaremba for Oilprice.com

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